

Chapter 4: Maternal Physiology — Revision Notes

Williams Obstetrics 26e, pp. 146–227 of the PDF. High-yield numbers are in **bold**. The six tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 19 figures are included from the PDF.

Big picture: Almost every organ system adapts, mostly driven by the fetus and placenta, and nearly all return to baseline after delivery and lactation. Normal changes can mimic disease or unmask pre-existing disease.

1. Reproductive Tract

Uterus

Parameter	Nonpregnant	Term
Weight	70 g	~1100 g
Cavity volume	≤10 mL	~5 L (may be ≥20 L) — 500–1000× increase
Myometrial thickness	—	1–2 cm at term

- Early growth = **hypertrophy** (estrogen ± progesterone), which is why it also happens with ectopic pregnancy. After **12 wk**, growth is mainly from **pressure of the conceptus**. Few new myocytes form.
- Growth is greatest at the **fundus** and around the placental site.
- **Three muscle layers:** outer hood-like; **middle interlacing "figure-of-8" layer** (constricts vessels after delivery → hemostasis); inner sphincter-like layer at the tubal ostia and internal os.
- **Shape:** pear → globular by 12 wk → ovoid. It **leaves the pelvis at 12 wk**.
- **Dextrorotation** (the rectosigmoid is on the left).
- **Supine** → the uterus rests on the vertebral column and great vessels.

Braxton Hicks contractions

- 5–25 mm Hg, irregular.
- Near term they occur every 10–20 min → **false labor**.
- Electrical synchrony develops **2× faster in multiparas**.

Uteroplacental blood flow

- **~450 mL/min** midtrimester → **500–750 mL/min at 36 wk**.
- Contractions reduce flow in proportion to their intensity; a **tetanic contraction causes a precipitous fall**.
- FGR fetuses tolerate labor poorly because their baseline flow is already reduced.

Regulation

- Flow ∝ **radius⁴**. Uterine artery 3.3 → 3.7 mm = 50% ↑ velocity.
- **Spiral arteries** dilate and **lose contractility** because of trophoblast invasion.
- **Nitric oxide / eNOS** is the key mediator, upregulated by shear stress, estrogen, progesterone, activin, **PIGF**, **VEGF**.
- **sFlt-1** binds PIGF/VEGF → **preeclampsia** (Ch 40).
- **Refractoriness to angiotensin II** raises uteroplacental flow.
- Also: relaxin; adipocytokines (chemerin, visfatin, leptin, resistin, adiponectin); **miR-17–92 and miR-34** (spiral artery remodeling).

Cervix

- Softens and turns **bluish at 1 month** (vascularity, edema, collagen remodeling, gland hyperplasia). By term, the **glands make up ½ of cervical mass**.
- **Eversion** (columnar epithelium on the portio) → red, velvety, bleeds easily (e.g., on Pap smear).
- **Mucus plug** (rich in IgA/IgG and cytokines) → expelled as **"bloody show"**.
- Mucus pattern: **beading** (progesterone, poor crystallization) is normal. **Ferning** suggests **amnionic fluid leak**.
- **Arias-Stella reaction**: endocervical gland hyperplasia/hypersecretion that mimics atypical glandular cells on Pap.



Figure 4-1. Cervical eversion of pregnancy (colposcopic view): columnar epithelium on the portio.

Ovaries

- Ovulation stops.
- **Corpus luteum** is maximal for **6–7 wk** (4–5 wk post-ovulation). Removal **before 7 wk** → progesterone falls → abortion, so give **progesterone support**. After 7 wk, removal is usually safe.
- **Decidual reaction** on the ovary/serosa: red blisters that bleed easily (seen at cesarean).
- Ovarian vascular pedicle **0.9 → 2.6 cm**.
- **Relaxin** comes from the corpus luteum, decidua and placenta (same pattern as hCG).
 - It remodels connective tissue, ↑ renal hemodynamics, ↓ osmolality and ↑ arterial compliance.
 - ⚠ It does **NOT** cause peripheral joint laxity or pelvic girdle pain.
- **Theca-lutein cysts (hyperreactio luteinalis)**:
 - Bilateral and linked to **high hCG**: GTD, multifetal pregnancy, diabetes, anti-D alloimmunization.
 - Associated with preeclampsia and hyperthyroidism.
 - **Maternal virilization in up to 30%**; the fetus is rarely virilized.
 - Self-limited: resolves after delivery.

Fallopian tubes

- Little hypertrophy, flattened epithelium, decidual cells but no continuous decidua.
- Rare torsion (usually with a paratubal or ovarian cyst).

Vagina & perineum

- **Chadwick sign** = violet color from ↑ vascularity.
- Thick white discharge with **pH 3.5–6**. The acidity comes from *Lactobacillus acidophilus* making lactic acid from glycogen.

- ↑ **Candidiasis** risk, especially in the 2nd–3rd trimesters.
- The vaginal wall thickens, connective tissue loosens and smooth muscle hypertrophies.

Pelvic organ prolapse and incontinence

- The uterus may **incarcerate at 10–14 wk** → replace it and use a pessary.
- **Urinary incontinence: ~20% in the 1st trimester → ~40% in the 3rd**, mostly **stress UI**.
 - Risk factors: age >30, obesity, smoking, constipation, GDM.
- Cystocele/rectocele may rarely block fetal descent → empty the bladder or rectum and reduce the bulge.

2. Breasts

- Tenderness early; they enlarge after **2 months**. Visible veins and larger, darker nipples.
- **Colostrum** after the first few months.
- **Montgomery glands** = hypertrophic sebaceous glands in the areola.
- **Gigantomastia** (rare) → bromocriptine, then later surgical reduction.
- Breast size does **not** correlate with milk production.

3. Skin

- **89%** of women have at least one physiological skin change.
- **Striae gravidarum:**
 - 70% abdomen, 41% hips/thighs, 33% breasts.
 - Risk factors: young age, family history, weight and weight gain.
 - Aloe vera and almond oil ↓ itching.
- **Diastasis recti:** if severe → ventral hernia.
- **Hyperpigmentation (up to 90%):**
 - **Linea nigra; chloasma/melasma gravidarum** ("mask of pregnancy"); areolae and genital skin.
 - Cause: MSH, estrogen, progesterone.
- **Vascular spiders and palmar erythema** (hyperestrogenemia): no clinical significance, and they regress.
- **Hair:** anagen is prolonged in pregnancy. Postpartum hair shedding is **telogen effluvium**.

4. Metabolic Changes

- **BMR ↑ 20%** by the 3rd trimester (+10% more with twins).
- **Extra energy needed: 85 / 285 / 475 kcal/d** in the 1st / 2nd / 3rd trimesters.
- Average **weight gain 12.5 kg (27.5 lb)**.

Table 4-1. Additional Energy Demands During Normal Pregnancy

Rates of tissue deposition

	1st Tri (g/d)	2nd Tri (g/d)	3rd Tri (g/d)	Total (g/280 d)
Weight gain	17	60	54	12,000
Protein deposition	0	1.3	5.1	597
Fat deposition	5.2	18.9	16.9	3741

Energy cost of pregnancy (estimated from BMR and energy deposition)

	1st Tri (kJ/d)	2nd Tri (kJ/d)	3rd Tri (kJ/d)	Total MJ	Total kcal
Protein deposition	0	30	121	14.1	3370
Fat deposition	202	732	654	144.8	34,600
Efficiency of energy utilization*	20	76	77	15.9	3800
Basal metabolic rate	199	397	993	147.8	35,130
Total energy cost	421	1235	1845	322.6	77,100

Assumes an average gestational weight gain of 12 kg. *Efficiency of food energy utilization for protein and fat deposition estimated as 0.90.

Table 4-2. Weight Gain Based on Pregnancy-Related Components (cumulative, g)

Tissues and Fluids	10 wk	20 wk	30 wk	40 wk
Fetus	5	300	1500	3400
Placenta	20	170	430	650
Amnionic fluid	30	350	750	800
Uterus	140	320	600	970
Breasts	45	180	360	405
Blood	100	600	1300	1450
Extravascular fluid	0	30	80	1480
Maternal stores (fat)	310	2050	3480	3345
Total	650	4000	8500	12,500

Water

- **Plasma osmolality ↓ 10 mOsm/kg**, early in pregnancy: reset of the **thirst and vasopressin thresholds** (relaxin plays a role).
- Extra water: **~6.5 L** in total (3.5 L fetus/placenta/fluid + 3.0 L maternal blood, uterus and breasts), about 15 lb.
- **Ankle pitting edema is normal** (~1 L). Causes: partial vena cava occlusion ↑ venous pressure, plus ↓ interstitial colloid osmotic pressure.

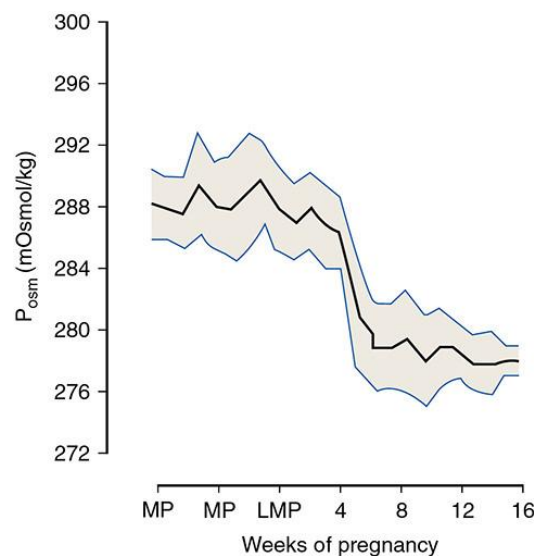


Figure 4-2. Plasma osmolality falls ~10 mOsm/kg early in pregnancy (reset of thirst and vasopressin thresholds).

Protein

- **~1000 g** protein gained: 500 g in the fetus + placenta, 500 g in the uterus, breasts and blood.

- Amino acids are higher on the fetal side (active placental transport).
- Estimated need is **1.22 g/kg/d early** and **1.52 g/kg/d late**, vs the current recommendation of **0.88 g/kg/d**, which may be too low.

Carbohydrate

- The triad: **mild fasting hypoglycemia + postprandial hyperglycemia + hyperinsulinemia**.
- **Peripheral insulin resistance**: insulin sensitivity is **30–70% lower** in late pregnancy.
 - Cause: progesterone, **placental GH**, prolactin, cortisol, TNF, leptin.
 - Effect: a sustained glucose supply to the fetus.
- Insulin half-life is **unchanged**.
- **↑ Hepatic gluconeogenesis** in the 3rd trimester.
- **"Accelerated starvation"**: with fasting, fuel switches to lipids. Prolonged fasting → rapid **ketonemia**.

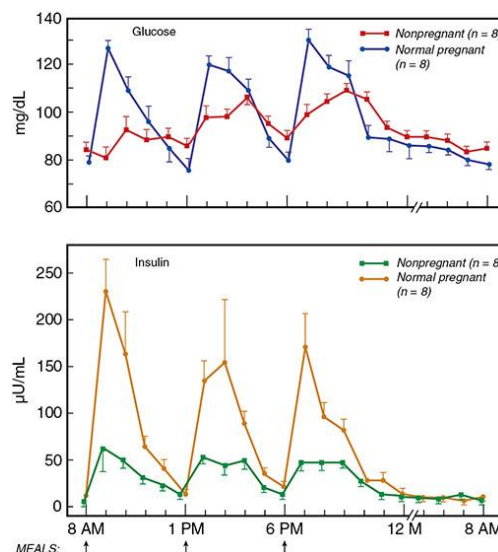


Figure 4-3. Diurnal glucose and insulin in late pregnancy: fasting hypoglycemia, postprandial hyperglycemia, hyperinsulinemia.

Fat

- **Anabolic** in the 1st–2nd trimesters (fat storage) → **catabolic** in the 3rd (lipolysis ↑, lipoprotein lipase ↓). This spares glucose and amino acids for the fetus.
- Maternal **hyperlipidemia** is caused by insulin resistance and estrogen.
- Endothelial function actually **improves**, because HDL ↑ inhibits LDL oxidation.
- Breastfeeding ↓ triglycerides and ↑ HDL.

Table 4-3. Plasma Concentrations of Lipids

Lipid	Nonpregnant	Third Trimester (mean ± SD)
Total cholesterol	<200 mg/dL	267 ± 30 mg/dL
LDL	<100 mg/dL	136 ± 33 mg/dL
HDL	40–60 mg/dL	81 ± 17 mg/dL
Triglycerides	<150 mg/dL	245 ± 73 mg/dL

Leptin and other adipocytokines

- **Leptin**: peaks in the 2nd trimester and plateaus at **2–4× nonpregnant** levels; the placenta is a major source, so it falls after delivery.
 - "Leptin resistance" promotes energy storage.
 - High levels are linked to preeclampsia, GDM and fetal distress.

- Low fetal leptin → FGR.
- **Adiponectin:** from maternal fat, **not the placenta**. An insulin sensitizer that falls in GDM, but it is **not useful for predicting GDM**.
- **Ghrelin:** from the stomach and placenta; a role in fetal growth.
- **Visfatin + leptin** ↑ → impaired uterine contractility (may explain dysfunctional labor in obesity).

Electrolytes and minerals

- Retained: **~1000 mEq Na⁺ and 300 mEq K⁺**, yet serum levels are **slightly** ↓.
- **Calcium:**
 - **Total Ca** ↓ (because albumin ↓); **ionized Ca unchanged**.
 - The fetus takes **~30 g Ca** by term, **80% in the 3rd trimester**.
 - Met by **doubled intestinal absorption** via **1,25(OH)₂D₃ (doubled)**, driven by **PTH-rP**.
- **Magnesium** ↓ (total and ionized). Phosphate is normal.
- **Iodine needs** ↑:
 1. ↑ maternal T4 production
 2. Fetal thyroid hormone production in the 2nd half
 3. **Renal iodine clearance** ↑ **30–50%** - Excess iodine → fetal hypothyroidism (**Wolff-Chaikoff effect**).

5. Hematological Changes

Blood volume

- ↑ **40–45%** after **32–34 wk**. Plasma ↑ ~15% by 12 wk. Fastest rise is in the midtrimester, then a plateau near term. Greater with twins.
- It happens even with a **hydatidiform mole**, so a fetus isn't needed.
- **Functions:**
 1. Meets the needs of the enlarged uterus
 2. Supplies nutrients to the fetus and placenta
 3. Protects against impaired venous return (supine or erect)
 4. **Protects against blood loss at delivery**
- **RBC mass** ↑ **~450 mL**, but plasma ↑ more → **dilutional** ↓ **Hb/Hct** and ↓ viscosity. Mild erythroid hyperplasia and ↑ reticulocytes (EPO ↑).
- **Hb at term averages 12.5 g/dL**. **<11.0 g/dL = abnormal**, usually iron deficiency.

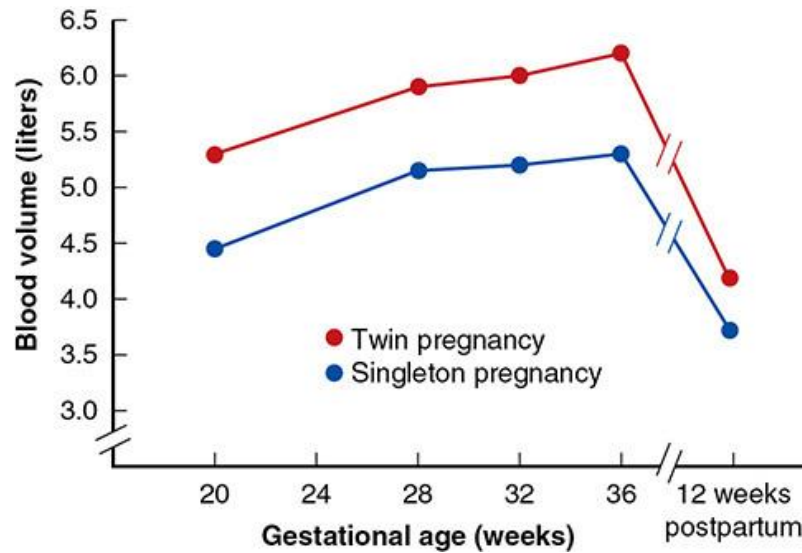


Figure 4-4. Blood volume expansion: singletons vs twins; plateau near term, falls postpartum.

Iron

- Women's total body iron is **2.0–2.5 g**, with stores of only **~300 mg**.
- **Hepcidin ↓ early** → ↑ absorption via ferroportin and ↑ transfer to the fetus.

Iron requirement ≈ 1000 mg

Use	Amount
Fetus + placenta	300 mg
Obligatory losses (mainly GI)	200 mg
RBC mass expansion (450 mL × 1.1 mg/mL)	500 mg

- Most of it is needed in the **second half: 6–7 mg/d**, which diet and stores can't supply → **supplement**.
- The **fetus gets iron even if the mother is severely anemic** (maternal Hb 3 vs fetal Hb 16 g/dL).
- Blood loss with vaginal delivery is **500–600 mL**. The leftover hemoglobin iron becomes stores.

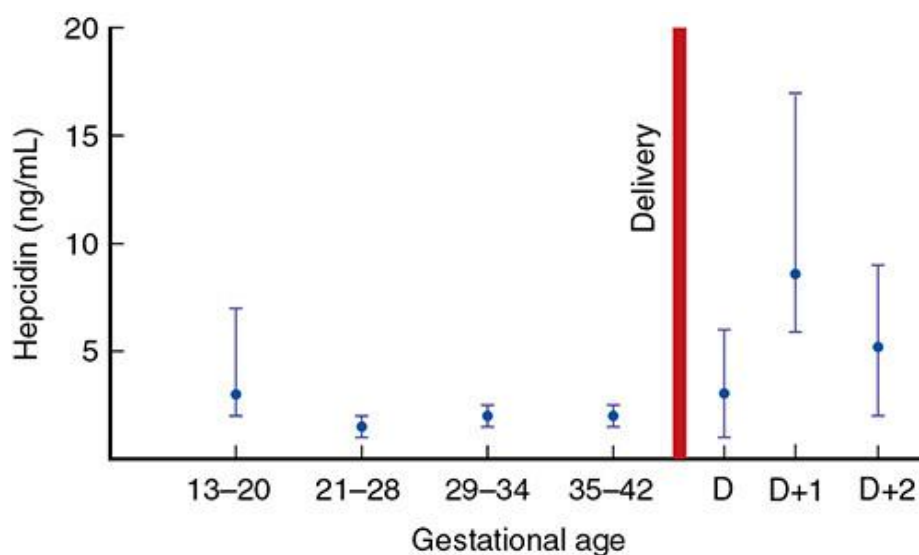


Figure 4-5. Hepcidin falls early in pregnancy (↑ iron absorption) and rises after delivery.

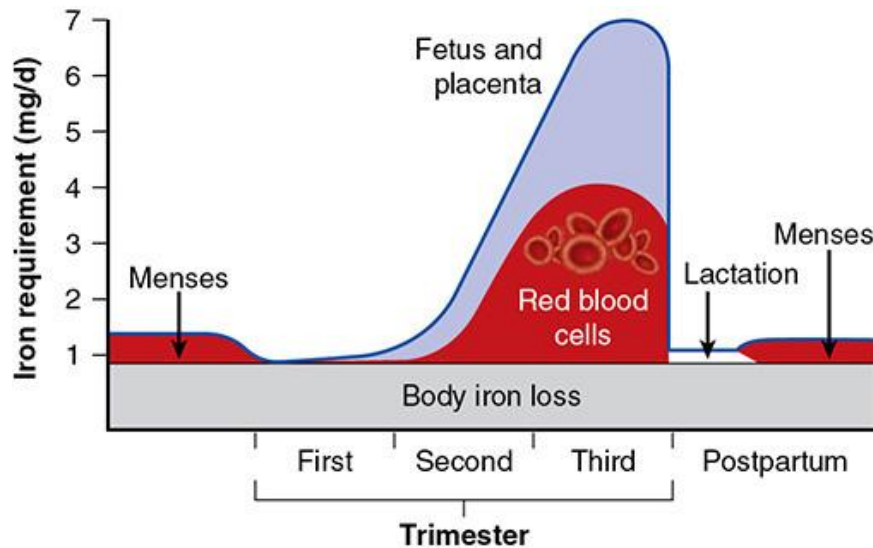


Figure 4-6. Daily iron requirement across pregnancy (55-kg woman): most is needed in the second half.

Immunology

- **Th1 → Th2 shift.** Suppression of Th1/Tc1 → ↓ IL-2, IFN, TNF.
- Special **MHC molecules on trophoblast.**
- Cervical mucus IgA/IgG ↑.
- **IgG crosses to the fetus in the 3rd trimester** (passive immunity).

Leukocytes and inflammatory markers

- **WBC up to 15,000/μL**, and **≥25,000/μL in labor and the early puerperium.**
- T lymphocytes ↑ (relative), B cells unchanged, CD4:CD8 ratio unchanged.
- **Leukocyte alkaline phosphatase ↑.**
- **CRP ↑**, but 95% of non-laboring women have **≤1.5 mg/dL.**
- **ESR ↑** (globulin + fibrinogen).
- **C3, C4 ↑.**
- **Procalcitonin** is low midpregnancy and ↑ at term and postpartum. It is a **poor predictor of chorioamnionitis** after PPROM.

Coagulation

- **Hypercoagulable (procoagulant) state:**
 - **All clotting factors ↑ except XI and XIII; XIII ↓.**
 - Thrombin generation ↑; fibrinolysis is probably ↓.
- **Fibrinogen: 300 → 450 mg/dL** (range 300–600), ↑ ~50%.
- **D-dimer ↑**, so it is less useful for diagnosing VTE.
- **Regulatory proteins:**
 - **Activated protein C resistance ↑** (APC 2.4 → 1.9 U/mL).
 - **Free protein S ↓** (0.4 → 0.16 U/mL).
 - **Antithrombin ↓ 13%** by term and ↓ 30% more by 12 h postpartum; back to baseline by 72 h.

Table 4-4. Coagulation Factor Normal Values Across Pregnancy

Factor	Nonpregnant	1st Tri	2nd Tri	3rd Tri
Fibrinogen (mg/dL)	233–496	244–510	291–538	373–619*
D-dimer (µg/mL)	0.22–0.74	0.05–0.95	0.32–1.29	0.13–1.7
Factor V (% activity)	50–150	75–95	72–96	60–88
Factor VII (% activity)	50–150	100–146	95–153	149–211
Factor VIII (% activity)	50–150	90–110	97–312	143–353
Protein C, functional (%)	70–130	78–121	83–133	67–135
Protein S, functional (%)	65–140	57–95	42–68	16–42

The printed table reads "373–6.19", an obvious typo for 373–619.

Platelets and spleen

- **Platelets** ↓ across pregnancy (lower in twins), and normal again by **4–12 wk postpartum**.
 - Causes: hemodilution, ↑ consumption (larger, younger platelets), splenic sequestration.
- **Spleen** ↑ **up to 50%** vs the 1st trimester (68% vs nonpregnant).

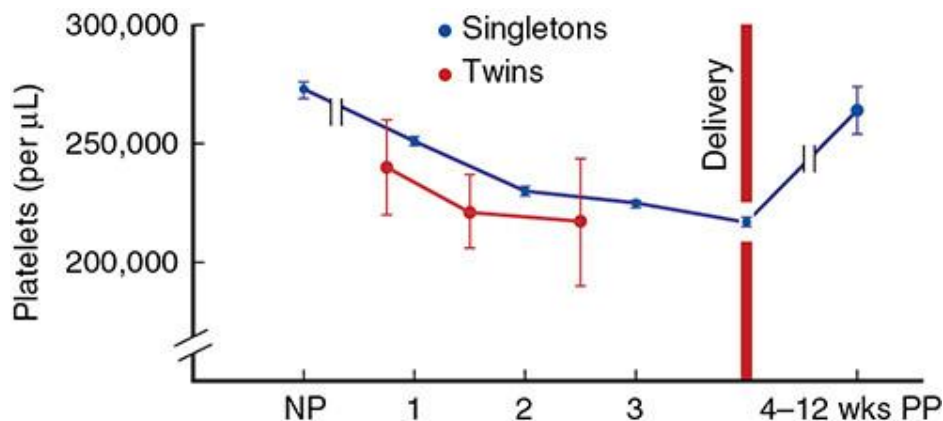


Figure 4-7. Platelet counts fall across pregnancy (lower with twins) and recover by 4–12 wk postpartum.

6. Cardiovascular System

- Changes start by **8 weeks**. **Cardiac output** ↑ **from week 5** (↓ SVR + ↑ HR).
- **Resting pulse** ↑ ~10 bpm.
- **Heart:**
 - Displaced **up and to the left and rotated** → larger silhouette on chest X-ray. A benign pericardial effusion is common.
 - **LV mass** ↑ **30–35%** (concentric remodeling from 26–30 wk; resolves within 3 months).
 - Ejection fraction and septal thickness are unchanged.
 - Cardiac function is **edynamic** (normal on the Braunwald graph).
- **ECG:** **slight left-axis deviation**; Q waves in II, III, aVF; flat or inverted T waves in III and V1–V3.
- **Heart sounds:**
 - **Exaggerated split S1** (both components louder)
 - No definite change in S2
 - **Loud S3**
 - **Systolic murmur is common**
 - Soft diastolic murmur (transient) and a mammary souffle (continuous) are possible

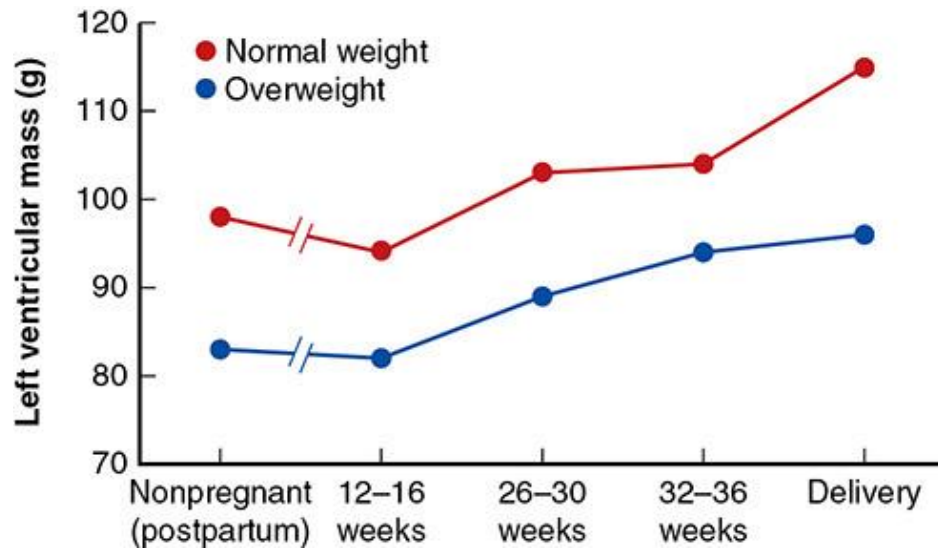


Figure 4-9. Left ventricular mass rises from 26–30 wk in normal-weight and overweight women.

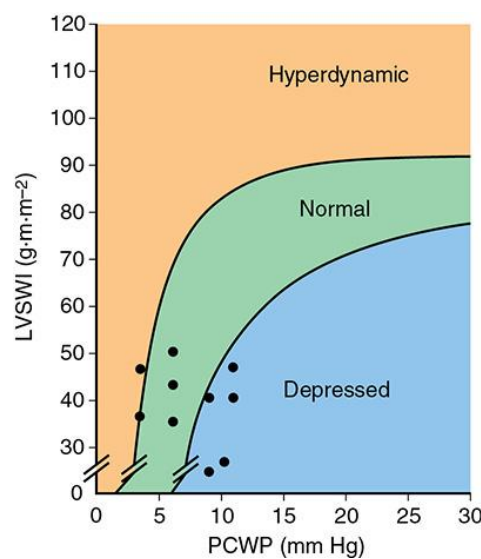


Figure 4-10. Braunwald graph (LVSWI vs PCWP): cardiac function in late pregnancy is normal (eudynamic).

Cardiac output

- Highest in the **lateral recumbent** position.
- **Supine** → **aortocaval compression** → ↓ CO. Rolling to the left side ↑ CO by **~20% at 26–30 wk** and **10% at 32–34 wk**. Fetal O₂ saturation is **10% higher** with the mother in the lateral position during labor.
- **Twins**: a further **~20% ↑**; cardiovascular reserve is reduced.
- CO ↑ moderately in the **1st stage** of labor and **appreciably in the 2nd stage**.

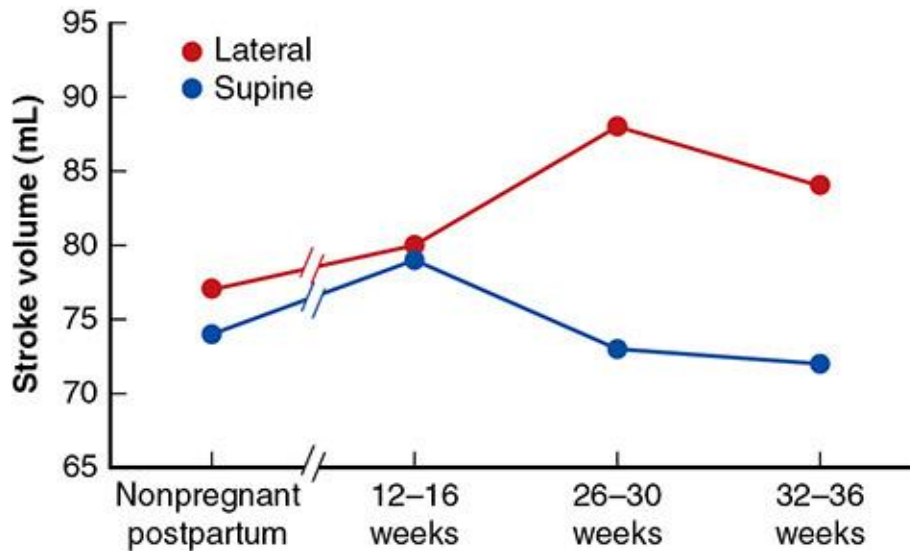


Figure 4-8. Stroke volume, lateral vs supine: supine position lowers stroke volume in late pregnancy.

Table 4-5. Central Hemodynamic Changes in 10 Normal Nulliparous Women Near Term and Postpartum (Clark, 1989)

Parameter	Pregnant (35–38 wk) ^a	Postpartum (11–13 wk)	Change ^b
Mean arterial pressure (mm Hg)	90 ± 6	86 ± 8	NSC
Pulmonary capillary wedge pressure (mm Hg)	8 ± 2	6 ± 2	NSC
Central venous pressure (mm Hg)	4 ± 3	4 ± 3	NSC
Heart rate (beats/min)	83 ± 10	71 ± 10	+17%
Cardiac output (L/min)	6.2 ± 1.0	4.3 ± 0.9	+43%
Systemic vascular resistance (dyn/sec/cm ⁻⁵)	1210 ± 266	1530 ± 520	-21%
Pulmonary vascular resistance (dyn/sec/cm ⁻⁵)	78 ± 22	119 ± 47	-34%
Serum colloid osmotic pressure (mm Hg)	18.0 ± 1.5	20.8 ± 1.0	-14%
COP–PCWP gradient (mm Hg)	10.5 ± 2.7	14.5 ± 2.5	-28%
LV stroke work index (g/m/m ²)	48 ± 6	41 ± 8	NSC

^aMeasured in lateral recumbent position. ^bSignificant unless NSC = no significant change. COP = colloid osmotic pressure; PCWP = pulmonary capillary wedge pressure.

Memory hook: "Output and rate go UP, resistance and COP go DOWN, pressures (MAP, PCWP, CVP) and LVSWI stay the SAME." Pregnancy is **not** a continuous "high-output" state.

The ↓ COP–PCWP gradient helps explain why pregnant women are more prone to **pulmonary edema**.

Blood pressure and venous system

- **BP nadir at 24–26 wk**, then it rises. **Diastolic falls more than systolic**.
- **Supine hypotensive syndrome** in ~10% (compression of the great vessels).
- **Femoral venous pressure** (supine): **8 → 24 mm Hg** at term. Antecubital pressure is unchanged.
 - Consequences: dependent edema, varicosities, hemorrhoids, **DVT risk**.

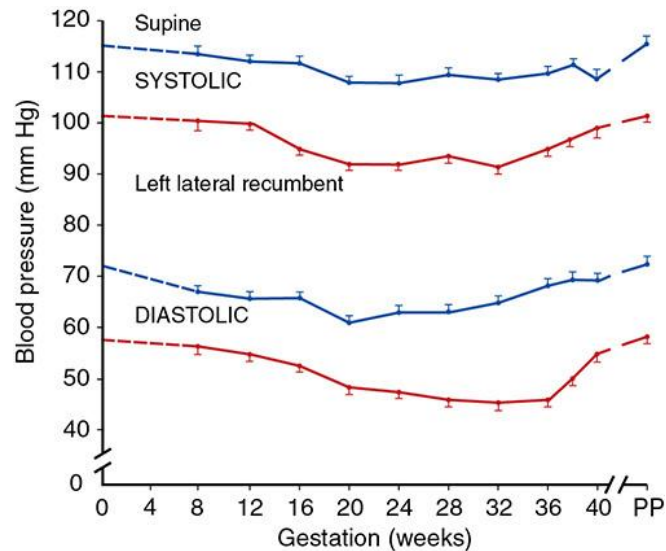


Figure 4-11. Blood pressure (supine vs left lateral): nadir at 24–26 wk; diastolic falls more than systolic.

Vasoactive systems

System	Change in pregnancy
Renin–angiotensin–aldosterone	All components ↑. Refractory to angiotensin II (progesterone, PIGF). Losing refractoriness predicts hypertension (Gant, 1973); it is lost within 15–30 min after the placenta delivers.
ANP / BNP	Stay in the nonpregnant range ; BNP <20 pg/mL. ↑ in severe preeclampsia.
Prostaglandins	↑ PGE ₂ (renal, natriuretic); ↑ PGI ₂ (vasodilation). The PGI ₂ :thromboxane ratio matters in preeclampsia.
Endothelin-1	Vasoconstrictor. Sensitivity is unchanged in normal pregnancy; ↑ in preeclampsia.
Nitric oxide	Vasodilator; placental vascular tone. Abnormal in preeclampsia.

7. Respiratory Tract

- **Diaphragm rises ~4 cm.** Subcostal angle widens; transverse chest diameter ↑ 2 cm; circumference ↑ 6 cm.
- Diaphragmatic excursion **increases**. Dyspnea is common.

Volume / capacity	Change
Functional residual capacity (FRC)	↓ 20–30% (400–700 mL)
Expiratory reserve volume	↓ 15–20% (200–300 mL)
Residual volume	↓ 20–25% (200–400 mL)
Inspiratory capacity	↑ 5–10% (200–350 mL)
Tidal volume	↑
Minute ventilation	↑ (10.7 → 14.1 L/min)
Total lung capacity	Unchanged or ↓ <5%
Respiratory rate	Unchanged
Vital capacity, lung compliance	Unchanged
Peak expiratory flow, airway conductance	↑ (total pulmonary resistance ↓)

- **O₂ consumption ↑ ~20%** (↑ 10% more with twins; ↑ **40–60% in labor**). The arteriovenous O₂ difference ↓.
- **Physiological dyspnea:** progesterone ↑ respiratory drive → ↑ tidal volume → ↓ PaCO₂.

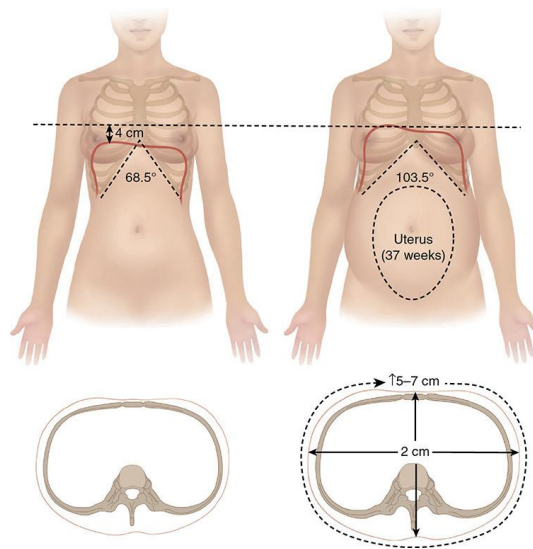


Figure 4-12. Chest wall changes: diaphragm $\uparrow$ 4 cm, subcostal angle $68.5^\circ \rightarrow 103.5^\circ$, transverse diameter $\uparrow$ 2 cm, circumference $\uparrow$ 5–7 cm.

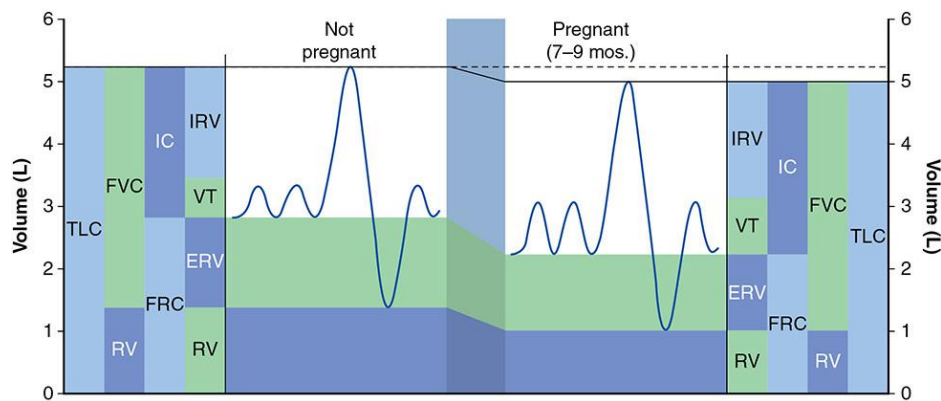


Figure 4-13. Lung volumes: FRC, ERV and RV fall; inspiratory capacity and tidal volume rise; TLC is about unchanged.

Acid-base

- **Compensated respiratory alkalosis.** Progesterone lowers the CO_2 threshold and raises chemoreflex sensitivity.
- **Bicarbonate $26 \rightarrow 22$ mmol/L.**
- pH $\uparrow$ slightly, shifting the O_2 dissociation curve **left** (Bohr effect). This is offset by $\uparrow$ **2,3-DPG**, which shifts it back **right**.
- Net result: $\downarrow$ PaCO_2 **helps CO_2 transfer from the fetus** and O_2 release to the fetus.

8. Urinary System

- **Kidney size $\uparrow$ ~1 cm.**
- **GFR:** $\uparrow$ 25% by 2 wk after conception and $\uparrow$ **50% by the start of the 2nd trimester**; stays up until term.
- **Renal plasma flow $\uparrow$ ~80%** by the end of the 1st trimester, then falls late in pregnancy.
 - Mechanism: hemodilution ($\downarrow$ oncotic pressure) + $\uparrow$ RPF. **Relaxin $\rightarrow$ NO $\rightarrow$ renal vasodilation.**
- Urinary frequency (60%) and **nocturia (80%)** in 3rd-trimester nulliparas.
- Supine position $\downarrow$ Na^+ excretion to less than half that of lateral recumbency.
- $\uparrow$ loss of amino acids and water-soluble vitamins.

Table 4-6. Renal Changes in Normal Pregnancy

Parameter	Alteration	Clinical relevance
Kidney size	~1 cm longer on radiograph	Returns to normal postpartum
Dilatation	Resembles hydronephrosis on sonogram or IVP (more marked on right)	Can be confused with obstructive uropathy. Retained urine → collection errors. Renal infections are more virulent. May cause "distention syndrome". Defer elective pyelography ≥12 wk postpartum.
Renal function	GFR and renal plasma flow ↑ ~50%	Serum creatinine ↓. >0.8 mg/dL (>72 μmol/L) is already borderline. Protein, amino acid and glucose excretion all ↑.
Acid-base	↓ bicarbonate threshold; progesterone stimulates the respiratory center	Serum HCO_3^- ↓ 4–5 mEq/L. PCO_2 ↓ 10 mm Hg. A PCO_2 of 40 mm Hg already represents CO_2 retention.
Plasma osmolality	Osmoregulation altered; thresholds for AVP release and thirst ↓; hormonal disposal rates ↑	Serum osmolality ↓ 10 mOsm/L (Na^+ ↓ ~5 mEq/L). Placental metabolism of AVP may cause transient diabetes insipidus.

AVP = vasopressin; IVP = intravenous pyelography.

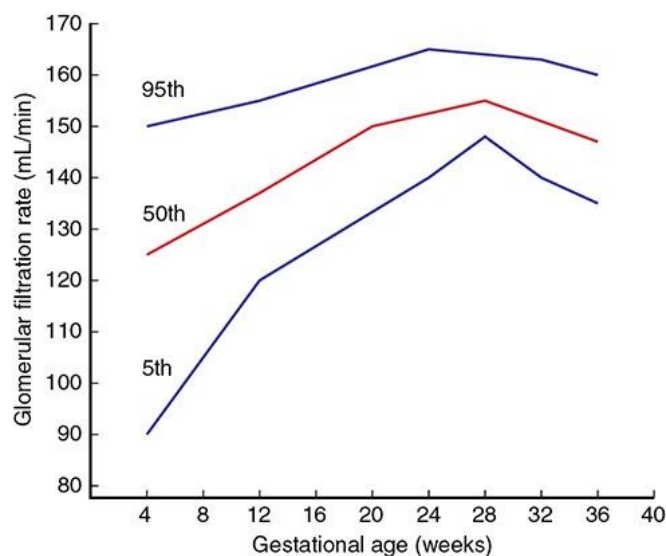


Figure 4-14. GFR (inulin clearance) across pregnancy: 5th, 50th and 95th percentiles.

Renal function tests

- **Creatinine 0.7 → 0.5 mg/dL. ≥0.9 mg/dL → look for renal disease.**
- Creatinine clearance is **~30% higher** than the nonpregnant 100–115 mL/min.
- Diurnal pattern reverses: water collects as edema by day and is mobilized at night → nocturia with dilute urine.

Urinalysis

- **Glucosuria** is often normal (↑ GFR + ↓ tubular reabsorption); **~1 in 6** women.
 - UK (NICE): test for GDM if dipstick is **2+ once or 1+ repeatedly.**
- **Proteinuria:**
 - Abnormal is **≥300 mg/d** (vs >150 mg/d nonpregnant).
 - Normal mean is **115 mg/d**; upper 95% limit **260 mg/d**; albumin 5–30 mg/d.
- **Measuring protein:**
 - Dipstick doesn't account for concentration.
 - 24-h collection has retention and timing errors from the dilated tract. Minimize them by hydrating and using lateral recumbency for 45–60 min at the start and end.
 - Protein:creatinine ratio is quick, but the thresholds vary. **Unreliable postpartum.**

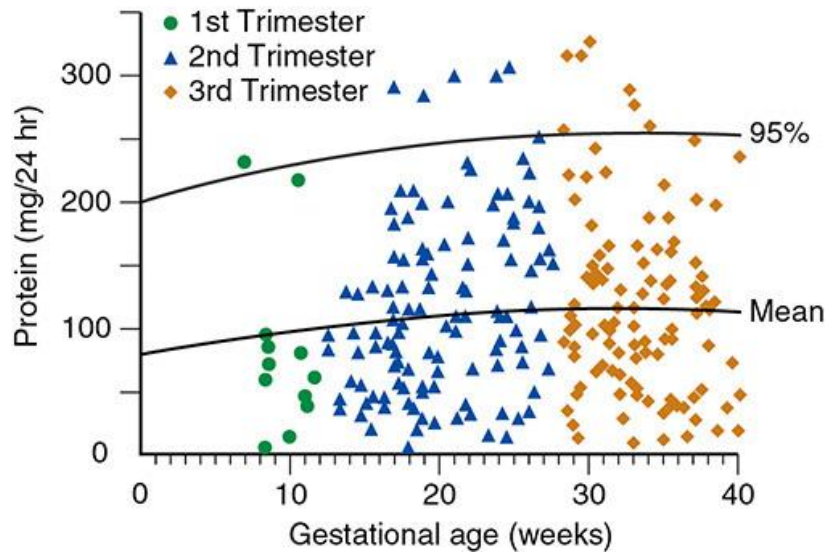


Figure 4-15. 24-hour urinary protein by gestational age: mean ~115 mg/d, upper 95% limit ~260 mg/d.

Ureters and bladder

- **Hydroureter/hydronephrosis, right > left (86% right-sided).** Why:
 1. The **sigmoid colon cushions** the left ureter
 2. The **dextrorotated** uterus compresses the right
 3. The dilated **right ovarian vein** crosses over the right ureter - Plus progesterone.
- Ureteral "kinks" are really gentle curves.
- Ureteroscopy complication rates are the same as when not pregnant.
- **Bladder:**
 - Trigone is elevated and widened. Pressure **8 → 20 cm H₂O**.
 - Urethral length ↑ (6.7 / 4.8 mm) and **intraurethral pressure 70 → 93 cm H₂O** preserve continence.
 - Still, **≥50% have some incontinence by the 3rd trimester** → always consider it when ruptured membranes are suspected.



Figure 4-16. IVP at 35 wk: moderate right (arrows) and mild left (arrowheads) hydronephrosis, both normal.

9. Gastrointestinal Tract

- The stomach and intestines are pushed **up**. The **appendix moves up and laterally** (can reach the right flank), so appendicitis presents differently.

- **Heartburn:** ↓ lower esophageal sphincter tone, ↑ intragastric and ↓ intraesophageal pressure, slower peristalsis.
- **Gastric emptying is unchanged in pregnancy**, but it is **prolonged in labor and after analgesics** → **aspiration risk with general anesthesia**.
- Hemorrhoids: constipation + ↑ rectal venous pressure.

Liver

- **Size unchanged.** Hepatic arterial and portal flow ↑. Stiffness ↑ in the 2nd–3rd trimesters.

Test	Change
Alkaline phosphatase	~2× ↑ (placental heat-stable isozyme)
AST, ALT, GGT, bilirubin	Slightly ↓
Albumin	↓ to ~3.0 g/dL (vs 4.3); total body albumin ↑
Globulin	Slightly ↑
Leucine aminopeptidase	Markedly ↑. Pregnancy-specific enzyme with oxytocinase/vasopressinase activity → transient DI

Pearl: a raised ALT/AST in pregnancy is **always abnormal**. A raised ALP alone is usually physiological.

Gallbladder

- **Volume ↑ 50%**, contractility ↓.
 - Cause: progesterone inhibits CCK-mediated contraction.
 - Result: stasis + lithogenic bile → **cholesterol stones** (~8% have sludge or stones).
- Intrahepatic cholestasis of pregnancy → see Ch 58.

10. Endocrine System

Pituitary

- **Enlarges ~135%** (lactotroph hyperplasia from estrogen). Up to 12 mm on MRI postpartum; back to normal by 6 months. Rarely compresses the chiasm → visual field loss.
- **Cell populations:**
 - Lactotrophs ↑
 - Gonadotrophs ↓
 - Corticotrophs and thyrotrophs unchanged
 - Somatotrophs suppressed (by placental GH)
- **The pituitary is not essential** for pregnancy: post-hypophysectomy pregnancies succeed with replacement.
- **Prolactinoma:** incidence not increased. A **macroadenoma (≥10 mm)** is more likely to grow.
- **Growth hormone:**
 - 1st trimester: maternal pituitary.
 - **Placental GH** detectable from 6 wk and **dominant by ~20 wk**. Differs by 13 amino acids; made by syncytiotrophoblast; acts via IGF-1.
 - Serum GH 3.5 → 14 ng/mL (plateau after 28 wk).
- **Prolactin:**
 - **~10× at term (~150 ng/mL)**. Paradoxically falls after delivery even with breastfeeding (pulsatile with suckling).
 - Essential for **lactation, not pregnancy**.
 - Amniotic fluid prolactin comes from the **decidua**, up to **10,000 ng/mL at 20–26 wk**.
 - A prolactin fragment is linked to **peripartum cardiomyopathy**.

- **Vasopressin (ADH) level is unchanged.**

Thyroid

- **hCG and TSH share an identical α -subunit.** hCG is weakly thyrotropic $\rightarrow$ **TSH $\downarrow$ in the 1st trimester** in >80% of women (still within the nonpregnant range).
- TRH is unchanged; it crosses the placenta.
- **TBG $\uparrow$ early, peaking at ~20 wk at 2 $\times$ baseline** (estrogen $\uparrow$ synthesis, $\downarrow$ clearance from sialylation).
 - $\rightarrow$ **Total T4 and T3 $\uparrow$** (rise from 6–9 wk, plateau at 18 wk).
 - $\rightarrow$ **Free T4 and T3 essentially unchanged**; free T4 peaks slightly with the hCG peak.
- Thyroid hormone production $\uparrow$ **40–100%**. Volume **12 $\rightarrow$ 15 mL**; **a goiter is not normal**, so evaluate it.
- $\sim 1/3$ of women have relative hypothyroxinemia with preferential T3 secretion.
- **Fetus:**
 - Fetal thyroid concentrates iodine from **10–12 wk**; its own TSH-driven secretion starts at **~20 wk**.
 - **~30% of cord T4 is maternal.**
- **Use gestational-age-specific TSH nomograms** (Dashe 2005). With nonpregnant cutoffs, **28%** of women with TSH above the 97.5th percentile would be missed.
- BMR $\uparrow$ up to 25%, mostly from the fetus.
- **Iodine:**
 - More than $1/3$ of the world is marginally deficient.
 - Severe deficiency $\rightarrow$ **cretinism** (>2 million people globally).

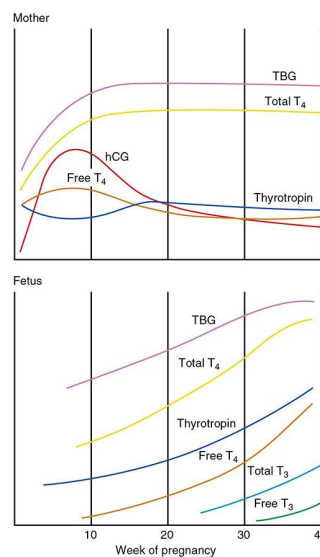


Figure 4-17. Maternal and fetal thyroid changes: TBG and total T4 rise; hCG transiently suppresses TSH; free T4 about unchanged.

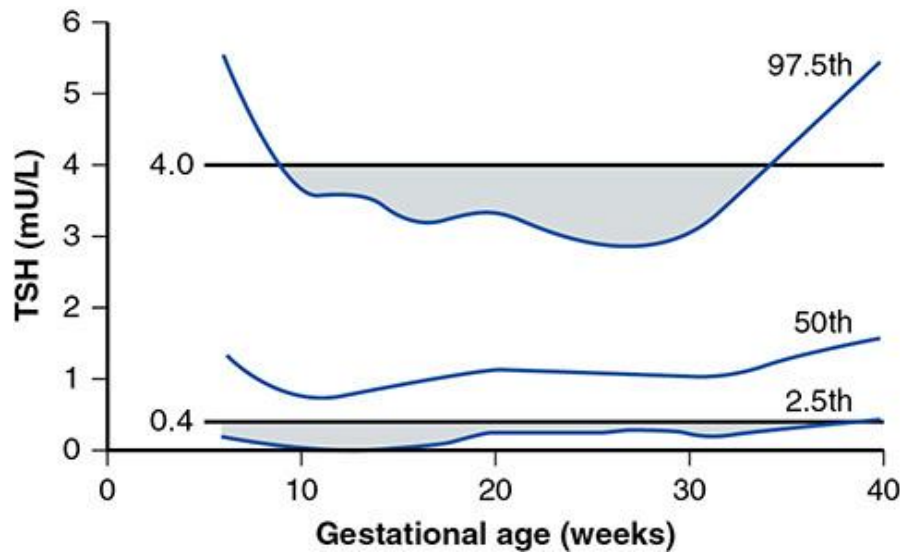


Figure 4-18. Gestational-age-specific TSH nomogram: nonpregnant cutoffs (0.4–4.0 mU/L) misclassify many women.

Parathyroid, calcium and bone

- **Calcitriol (1,25-(OH)₂D)** ↑ 2× → ↑ intestinal Ca absorption to **~400 mg/d** in the 3rd trimester.
- **PTH ↓ in the 1st trimester, then ↑**. Active vitamin D rises anyway because of **placental PTH / PTH-rP**.
- **PTH-rP ↑ markedly** (made by fetal tissues and the breasts).
- Fetal skeleton needs **30 g Ca** (3% of maternal skeletal Ca).
- Bone turnover markers stay ↑ up to 12 months postpartum → risk of **pregnancy-associated osteoporosis**.
- **Calcitonin** (thyroid C cells) protects the maternal skeleton. Fetal levels are ≥2× maternal.

Adrenal

Hormone	Change
Cortisol	Total ↑ (transcortin ↑). Free cortisol ↑. Secretion rate unchanged or ↓; half-life ~2× .
ACTH	↓ early, then ↑ in parallel with free cortisol (feedback reset at a higher set point; progesterone antagonism).
Aldosterone	↑ from 15 wk ; ~1 mg/d by the 3rd trimester (angiotensin II ↑). Protects against the natriuresis of progesterone and ANP.
Deoxycorticosterone	>15× ↑ (~1500 pg/mL). Made by the kidney (estrogen-stimulated), not the adrenal.
Androstenedione / testosterone	↑ (probably ovarian). The placenta aromatizes them to estradiol, so the fetus is protected even from androgen-secreting tumors.
DHEA-S	↓ (↑ hepatic 16α-hydroxylation + placental conversion to estrogen).

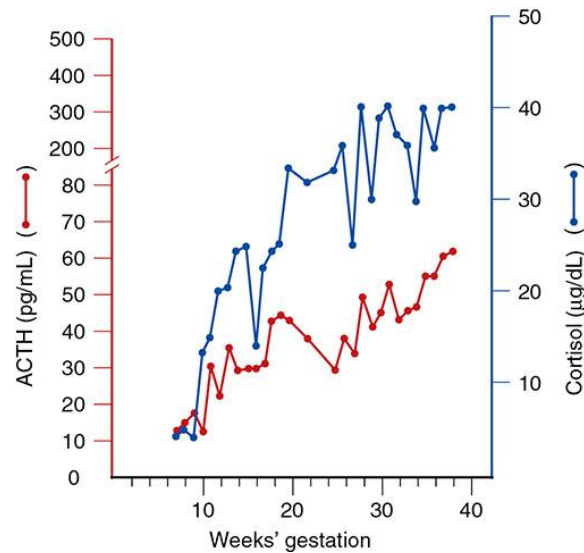


Figure 4-19. Serum cortisol and ACTH rise together across pregnancy.

11. Musculoskeletal System

- **Progressive lordosis** moves the center of gravity back over the legs.
- Sacroiliac, sacrococcygeal and pubic joint mobility ↑, mostly in the first half of pregnancy → low back pain.
 - ⚠ **Not** correlated with relaxin, estradiol or progesterone levels.
- **Symphyseal separation >1 cm** → significant pain.
- Neck flexion and shoulder slumping → ulnar/median nerve traction (can mimic **carpal tunnel syndrome**).
- Joints firm up again within **3–5 months** postpartum. Pelvic dimensions are unchanged by 3 months.

12. Central Nervous System

- **Cerebral blood flow ↓**:
 - MCA: 147 → 118 mL/min
 - PCA: 56 → 44 mL/min
 - **Autoregulation is unchanged.**
- **Memory**: a decline that is worst in the **3rd trimester**, transient and **not** explained by mood or sleep. Preeclampsia and eclampsia → lasting cognitive effects.
- **Eyes**:
 - **Intraocular pressure ↓**
 - Corneal sensitivity ↓
 - **Corneal thickness ↑** (contact lens intolerance)
 - **Krukenberg spindles**
 - Transient loss of accommodation
- **Sleep**: disrupted from **12 wk** to 2 months postpartum, worst postpartum. Linked to sleep-disordered breathing and postpartum blues or depression.

Quick-Recall: What Goes Up / Down / Stays the Same

↑ Increases	↓ Decreases	↔ Unchanged
Blood volume (40–45%), RBC mass, plasma volume	Hb/Hct (dilutional), blood viscosity	Respiratory rate
Cardiac output (40–50%), HR (+10–15 bpm), stroke volume	SVR, PVR, colloid osmotic pressure	MAP, PCWP, CVP, LVSWI
Tidal volume, minute ventilation, O ₂ consumption	FRC, ERV, RV, PaCO ₂ , HCO ₃ ⁻	Total lung capacity, vital capacity, lung compliance
GFR (50%), RPF, creatinine clearance	Serum creatinine, osmolality, Na ⁺ , K ⁺	Antecubital venous pressure
All clotting factors (except XI, XIII), fibrinogen, D-dimer, WBC, ESR, CRP	Platelets, protein S, antithrombin, factor XIII	CD4:CD8 ratio, B cells
ALP (placental), albumin <i>total</i> , lipids (TC, LDL, HDL, TG)	Albumin <i>concentration</i> , AST/ALT/GGT/bilirubin	Liver size, gastric emptying (before labor)
TBG, total T4/T3, prolactin, cortisol, ACTH, aldosterone, DOC, 1,25-D, PTH-rP	TSH (1st tri), DHEA-S, total Ca, Mg	Free T4/T3, ionized Ca, vasopressin level, insulin half-life
Insulin, insulin resistance, leptin	Fasting glucose, insulin sensitivity	Cerebral autoregulation

Key Numbers to Memorize

- Uterus **70 g → 1100 g**, **5 L** volume; uteroplacental flow **500–750 mL/min**
- Weight gain **12.5 kg**; extra water **6.5 L**; protein **1000 g**; iron **1000 mg** (6–7 mg/d late)
- Blood volume **+40–45%**; RBC **+450 mL**; Hb **<11 g/dL = anemia**
- WBC to **15,000** (labor **≥25,000**); fibrinogen **~450 mg/dL**
- Cardiac output **+43%**, HR **+17%**, SVR **-21%**; BP nadir **24–26 wk**; supine hypotension **10%**
- Diaphragm **↑4 cm**; FRC **↓20–30%**; HCO₃⁻ **22 mmol/L**; O₂ use **+20%**
- GFR **+50%**; creatinine **0.5 mg/dL** (≥0.9 abnormal); proteinuria **≥300 mg/d** abnormal
- Hydronephrosis **right-sided in 86%**
- Pituitary **+135%**; prolactin **~150 ng/mL**; thyroid volume **12 → 15 mL**; aldosterone **1 mg/d**; DOC **15×**